

Highly Reliable CMP-Free Ge-on-Si Waveguide Photodiodes for Data Center Optical Interconnects

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Abstract We demonstrate reliable CMP-free Ge-on-Si photodiodes, validated through extensive accelerated aging, showing stable dark current. The device exhibits 0.89 A/W responsivity, 65.3 GHz bandwidth, and sub-10 nA dark current, enabling simplified silicon photonics integration for optical interconnects in AI data centers with strict reliability requirements. ©2026 The Author(s)

Introduction

Silicon photonics has emerged as a key technology for next-generation optical interconnects, addressing the increasing bandwidth, energy-efficiency, and integration density demands of data center and artificial intelligence (AI) systems [1]. Ge-on-Si photodiodes (PDs) have become the preferred solution for near-infrared optical transceivers due to their CMOS compatibility, high absorption properties, and capability to achieve high responsivity and high bandwidth [2].

A conventional fabrication method of Ge-on-Si PDs relies on chemical-mechanical polishing (CMP) to planarize the top surface after Ge epitaxy [3, 4]. Although CMP enables tight control of the layer thickness and surface planarity, the process increases fabrication complexity and cost, and may induce surface damage or defects at the Ge-passivation interface [5]. These defects can act as generation-recombination centers [6], contributing to increased dark current and degraded device reliability [7, 8]. Recently, CMP-free Ge-on-Si integration schemes, which enable flexible engineering of Ge width and thickness while avoiding parasitic growth effects, have been implemented in imec's silicon photonics platform to achieve high performance PDs [9, 10].

In this paper, we report highly reliable CMP-free Ge-on-Si PDs fabricated on a 200-mm silicon photonics platform. We present a detailed study of the accelerated degradation of dark current under electrical stress. We demonstrate that CMP-free integration not only simplifies wafer-scale processing but also preserves excellent long-term stability.

Device description

O-band CMP-free Ge-on-Si PDs were fabricated in imec's 200-mm silicon photonics platform using a silicon-on-insulator (SOI) wafer with a 220 nm top Si layer on a thick buried oxide (BOX). A cross-sectional schematic of the device is shown in Fig. 1. The PDs exhibit a lateral p-i-n (LPIN) junction architecture, enabling the generation of

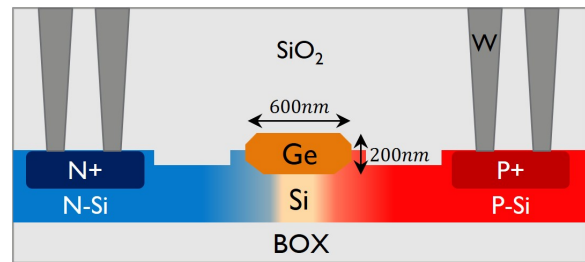


Fig. 1: Cross-section of the CMP-free Ge-on-Si photodiode.

high electric fields within the thin 200 nm Ge layer.

The Ge layer is formed by slow selective epitaxial growth into recessed Si region. The reduced deposition rate enables precise control over the Ge layer formation and thickness. To reduce threading dislocation density, a high-temperature post-growth annealing step is applied. The absence of a post-growth planarization step results in a simplified process flow with reduced risk of surface damage associated with mechanical polishing. In particular, an improvement of the top Ge/SiO₂ interface defectivity is expected to yield weaker interface-related leakage current and limited device-to-device variability.

Opto-electrical performance

In this study, 90 devices from a single wafer were electrically characterized at 25 °C. The distribution of dark current (I_{dark}) was plotted against the reverse voltage, as shown in Fig. 2(a). At a reverse bias of -2 V, a mean I_{dark} of 3.34 nA with a standard deviation of 0.55 nA is obtained. The narrow distribution demonstrates both the tight control of defectivity within the Ge layer and the wafer-scale manufacturability of the device.

Temperature-dependent I - V characteristics are used to identify the conduction mechanisms within the active regions of junction-based devices [11]. Fig. 2(b) presents the results for 32 devices selected from the initial population, measured from 25 °C to 175 °C.

The dominant transport mechanisms are quantitatively assessed using the effective activation energy (E_A), extracted from the Arrhenius relation

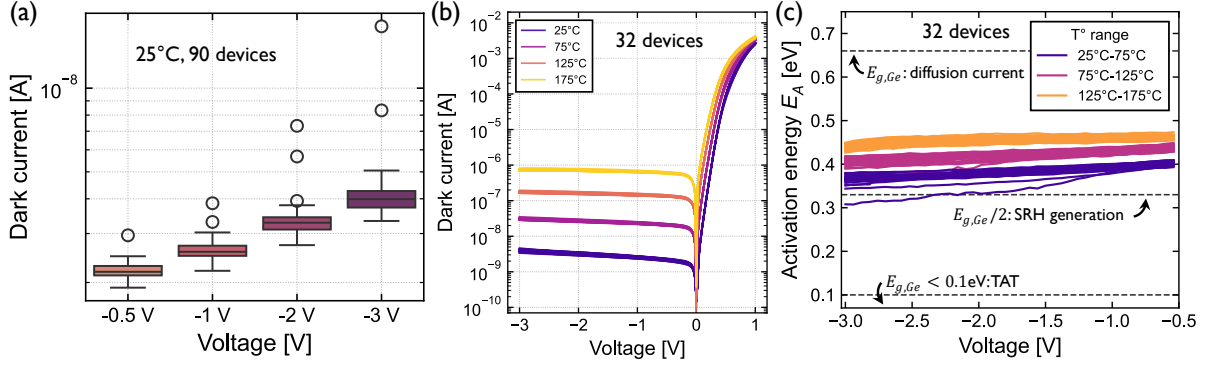


Fig. 2: (a) Wafer-scale distribution of I_{dark} at 25 °C. (b) Temperature-dependent I - V characteristics. (c) Voltage dependence of the effective activation energy (E_A) of I_{dark} extracted over three different temperature ranges.

for dark current [12]:

$$I_{dark} \propto \exp(-E_A/k_B T), \quad (1)$$

with k_B the Boltzmann constant and T the temperature. For each reverse bias, E_A is extracted over three different temperature ranges, and the resulting plots are shown in Fig. 2(c). The measured E_A values, ranging from 0.3 eV to 0.47 eV, indicate a mixed contribution from minority carrier diffusion and Shockley–Read–Hall (SRH) generation associated with defects within the depleted junction. The SRH contribution is more pronounced at lower temperatures and higher voltages. In contrast to thicker, CMP-based LPIN PDs previously reported in [8], the persistence of SRH-dominated I_{dark} at elevated temperatures highlights the impact of the compact CMP-free junction geometry, resulting in a strong local electric-field enhancement for a given bias. Importantly, almost no signature of trap-assisted tunneling (TAT) is observed ($0.1 \text{ eV} \leq E_A \leq 0.33 \text{ eV}$).

The device responsivity was measured at a wavelength of 1310 nm. The light was coupled into the PD through TE grating-coupler using a single mode fiber. Fig. 3(a) shows the voltage-dependent responsivity obtained from 48 devices across the wafer. At a reverse bias of -2 V, a high responsivity of $0.89 \pm 0.02 \text{ A/W}$ is achieved.

The high-speed performance of the PD was evaluated through opto-electrical frequency response measurements. The O-band 3-dB bandwidth (BW) was extracted from S_{21} responses of nine devices, as shown in Fig. 3(b). The measurements were performed using a 110 GHz Keysight network analyzer in combination with a 110 GHz external modulator from Thorlabs, with an input optical power of -11 dBm at the device, estimated by de-embedding the grating-coupler losses measured from reference waveguides. The BW increases with reverse bias, reaching a value of $65.3 \pm 1.5 \text{ GHz}$ at -2 V. Overall, good uniformity of performance is shown in these samples.

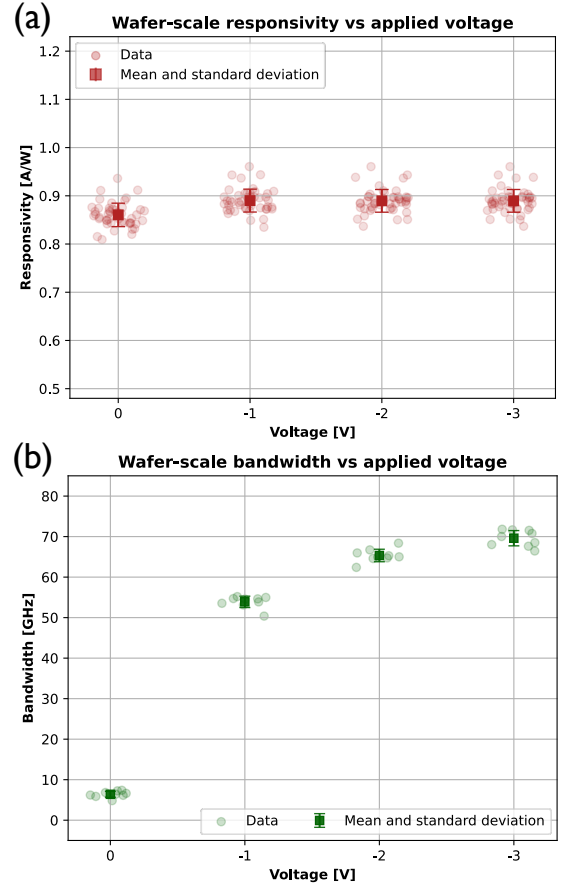


Fig. 3: (a) Responsivity and (b) 3-dB bandwidth of the O-band CMP-free Ge-on-Si PDs.

Wafer-level accelerated aging

Accelerated aging experiments were performed using constant voltage stress. High stress voltages (V_{stress}) ranging between -3 V to -5 V were applied at fixed temperatures of 150 °C and 175 °C. The I_{dark} was periodically monitored at -2 V during the short stress interruptions. The total stress duration was 10^4 s (2.8 h), followed by a relaxation sequence performed at 0 V for 0.9 h to evaluate potential recovery effects after stress removal [8].

As shown in Fig. 4, no significant degradation of dark current is observed, even at the highest V_{stress} of -5 V, corresponding to more than twice

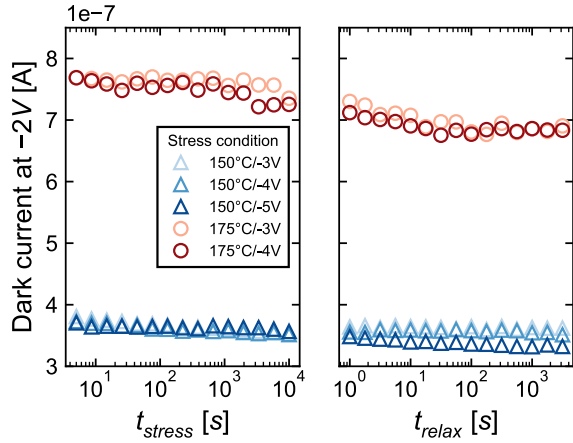


Fig. 4: Evolution of I_{dark} measured at -2 V during wafer-level accelerated aging and subsequent relaxation after stress removal, for various T_{stress} and V_{stress} combinations.

the typical operating bias of -2 V. These results contrast with CMP-based LPIN PDs, which have previously shown 10 % dark current increase under milder stress conditions of 125 °C and -3 V [8]. The enhanced stability reported here highlights the benefit of the CMP-free process, which limits defect formation near Ge/SiO₂ interfaces. As a result, charge trapping in pre-existing defects and the associated local electric-field enhancement [8] are effectively reduced.

High-temperature operating lifetime tests

To assess the long-term reliability of CMP-free PDs, packaged-level high-temperature operating lifetime (HTOL) experiments were performed at T_{stress} from 125 °C to 175 °C and V_{stress} of -2 V and -3 V. During the HTOL tests, the dark current evolution was continuously monitored under stress for up to 500 h, as shown in Fig. 5(a). The electrical performance was measured at standard operating condition (85 °C and -2 V) both before and after the HTOL sequence, as illustrated in Fig. 5(b). For each stress condition, five to six devices were tested to assess statistical repeatability.

Under most stress conditions, I_{dark} remained stable throughout aging. A modest increase, reaching at most a factor of two relative to the initial value, was only observed at the most severe condition of 175 °C and -3 V. The degradation rate is slightly accelerated at higher T_{stress} and V_{stress} , confirming thermally-assisted charge trapping effects enhanced by the electric field across the junction. The observed I_{dark} evolution exhibits a clear saturation behavior over time. This saturation suggests a finite number of available pre-existing defects that become progressively filled during stress [8].

Moreover, comparison of the dark current measured at 85 °C before and after 500 h of HTOL reveals no degradation and full recovery due to charge de-trapping effects after stress removal. These results illustrate the high tolerance of CMP-

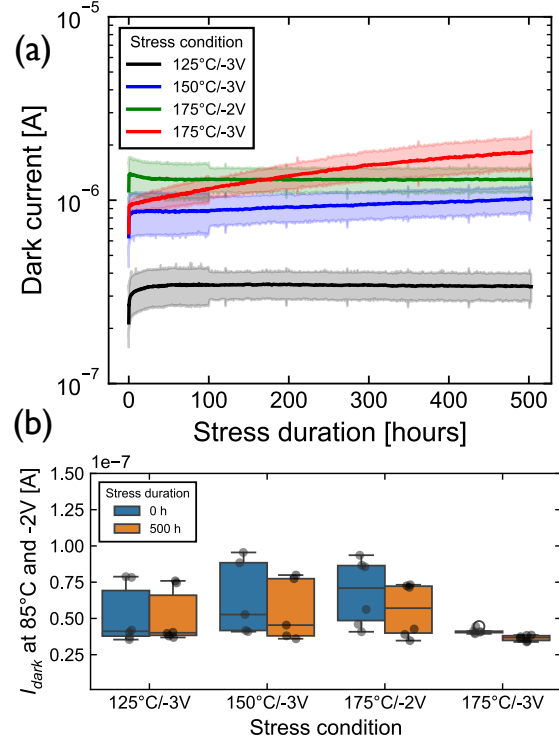


Fig. 5: (a) Mean dark current evolution over 500 h under four different stress conditions, showing limited degradation of I_{dark} . Each condition includes 5–6 stressed devices. The shaded areas represent the mean $\pm$ standard deviation across the device population. (b) Box plots comparing dark current before and after 500 h HTOL tests, measured at 85 °C and -2 V.

free Ge-on-Si photodiodes to permanent degradation during long-duration aging.

Note that the devices investigated in this work correspond to the baseline LPIN design. The deeply recessed variants enabling bandwidths beyond 100 GHz, reported in [9, 10], were not available at the time of this study and represent an important area for future reliability investigations.

Conclusions

We present a comprehensive reliability study of CMP-free Ge-on-Si photodiodes, exhibiting a high responsivity of 0.89 A/W, a 3-dB bandwidth of 65.3 GHz, and a low dark current below 10 nA. Across a broad range of electrical and thermal stress conditions, including wafer-level highly accelerated aging and 500 h HTOL testing, the dark current remains stable. The moderate increase in dark current observed under the most severe stress conditions is limited, saturates over time, and fully recovers. We show that CMP-free processing not only simplifies wafer-scale integration but also does not compromise the long-term photodiode stability by avoiding planarization-induced interface defects.

Acknowledgements

The authors would like to thank the imec's 200-mm silicon photonics platform, device integration and measurement teams.

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